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3110

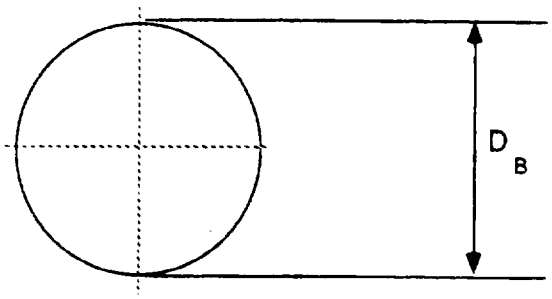


TABLE I. Basic diameter.

Nominal Ball Diameter	Basic Diameter (D_B)	Nominal Ball Diameter	Basic Diameter (D_B)	Nominal Ball Diameter	Basic Diameter (D_B)
	0.1240		0.2178		0.3115
	0.1242		0.2180		0.3117
	0.1244		0.2182		0.3119
	0.1246		0.2184		0.3121
	0.1248		0.2186		0.3123
1/8	0.1250	7/32	0.2188	5/16	0.3125
	0.1252		0.2190		0.3127
	0.1254		0.2192		0.3129
	0.1256		0.2194		0.3131
	0.1258		0.2196		0.3133
	0.1260		0.2198		0.3135
	0.1262		0.2200		0.3137
	0.1264		0.2202		0.3139
	0.1266		0.2204		0.3141
	0.1553		0.2490		0.3428
	0.1555		0.2492		0.3430
	0.1557		0.2494		0.3432
	0.1559		0.2496		0.3434
	0.1561		0.2498		0.3436
5/32	0.1563	1/4	0.2500	11/32	0.3438
	0.1565		0.2502		0.3440
	0.1567		0.2504		0.3442
	0.1569		0.2506		0.3444
	0.1571		0.2508		0.3446
	0.1573		0.2510		0.3448
	0.1575		0.2512		0.3450
	0.1577		0.2514		0.3452
	0.1579		0.2516		0.3454
	0.1865		0.2803		0.3740
	0.1867		0.2805		0.3742
	0.1869		0.2807		0.3744
	0.1871		0.2809		0.3746
	0.1873		0.2811		0.3748
3/16	0.1875	9/32	0.2813	3/8	0.3750
	0.1877		0.2815		0.3752
	0.1879		0.2817		0.3754
	0.1881		0.2819		0.3756
	0.1883		0.2821		0.3758
	0.1885		0.2823		0.3760
	0.1887		0.2825		0.3762
	0.1889		0.2827		0.3764
	0.1891		0.2829		0.3766

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REVIEWER ACTIVITIES:
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AIR FORCE - 71, 84
DLA - IS

MCSC N/A

P.A. OS Other cust AV, 99	International interest	Title Balls, Bearing, Grade 10, Aircraft-Quality Steel	MILITARY STANDARD MS3226
PROCUREMENT SPECIFICATION MIL-B-1083	SUPERSEDES MS3224 in part		PAGE 1 OF 7

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TABLE I. Basic diameter (continued).

Nominal Ball Diameter	Basic Diameter (D_B)	Nominal Ball Diameter	Basic Diameter (D_B)	Nominal Ball Diameter	Basic Diameter (D_B)
	0.4053		0.4990		0.5928
	0.4055		0.4992		0.5930
	0.4057		0.4994		0.5932
	0.4059		0.4996		0.5934
	0.4061		0.4998		0.5936
13/32	0.4063	1/2	0.5000	19/32	0.5938
	0.4065		0.5002		0.5940
	0.4067		0.5004		0.5942
	0.4069		0.5006		0.5944
	0.4071		0.5008		0.5946
	0.4073		0.5010		0.5948
	0.4075		0.5012		0.5950
	0.4077		0.5014		0.5952
	0.4079		0.5016		0.5954
	0.4365		0.5303		0.6240
	0.4367		0.5305		0.6242
	0.4369		0.5307		0.6244
	0.4371		0.5309		0.6246
	0.4373		0.5311		0.6248
7/16	0.4375	17/32	0.5313	5/8	0.6250
	0.4377		0.5315		0.6252
	0.4379		0.5317		0.6254
	0.4381		0.5319		0.6256
	0.4383		0.5321		0.6258
	0.4385		0.5323		0.6260
	0.4387		0.5325		0.6262
	0.4389		0.5327		0.6264
	0.4391		0.5329		0.6266
	0.4678		0.5615		0.6553
	0.4680		0.5617		0.6555
	0.4682		0.5619		0.6557
	0.4684		0.5621		0.6559
	0.4686		0.5623		0.6561
15/32	0.4688	9/16	0.5625	21/32	0.6563
	0.4690		0.5627		0.6565
	0.4692		0.5629		0.6567
	0.4694		0.5631		0.6569
	0.4696		0.5633		0.6571
	0.4698		0.5635		0.6573
	0.4700		0.5637		0.6575
	0.4702		0.5639		0.6577
	0.4704		0.5641		0.6579

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TABLE I. Basic diameter (continued).

Nominal Ball Diameter	Basic Diameter (D_B)	Nominal Ball Diameter	Basic Diameter (D_B)	Nominal Ball Diameter	Basic Diameter (D_B)
11/16	0.6865 0.6867 0.6869 0.6871 0.6873 0.6875 0.6877 0.6879 0.6881 0.6883 0.6885 0.6887 0.6889 0.6891	25/32	0.7803 0.7805 0.7807 0.7809 0.7811 0.7813 0.7815 0.7817 0.7819 0.7821 0.7823 0.7825 0.7827 0.7829 0.7831 0.7833 0.7835 0.7837	29/32	0.9053 0.9055 0.9057 0.9059 0.9061 0.9063 0.9065 0.9067 0.9069 0.9071 0.9073 0.9075 0.9077 0.9079 0.9081 0.9083 0.9085 0.9087
23/32	0.7178 0.7180 0.7182 0.7184 0.7186 0.7188 0.7190 0.7192 0.7194 0.7196 0.7198 0.7200 0.7202 0.7204	13/16	0.8115 0.8117 0.8119 0.8121 0.8123 0.8125 0.8127 0.8129 0.8131 0.8133 0.8135 0.8137 0.8139 0.8141 0.8143 0.8145 0.8147 0.8149	15/16	0.9365 0.9367 0.9369 0.9371 0.9373 0.9375 0.9377 0.9379 0.9381 0.9383 0.9385 0.9387 0.9389 0.9391 0.9393 0.9395 0.9397 0.9399
3/4	0.7490 0.7492 0.7494 0.7496 0.7498 0.7500 0.7502 0.7504 0.7506 0.7508 0.7510 0.7512 0.7514 0.7516 0.7518 0.7520 0.7522 0.7524	7/8	0.8740 0.8742 0.8744 0.8746 0.8748 0.8750 0.8752 0.8754 0.8756 0.8758 0.8760 0.8762 0.8764 0.8766 0.8768 0.8770 0.8772 0.8774	1	0.9990 0.9992 0.9994 0.9996 0.9998 1.0000 1.0002 1.0004 1.0006 1.0008 1.0010 1.0012 1.0014 1.0016 1.0018 1.0020 1.0022 1.0024

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P.A. OS Other cust AV, 99	International interest	Title Balls, Bearing, Grade 10, Aircraft-Quality Steel	MILITARY STANDARD MS3226
PROCUREMENT SPECIFICATION MIL-B-1083	SUPERSEDES MS3224 in part		PAGE 3 OF 7

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TABLE I. Basic diameter (continued).

Nominal Ball Diameter	Basic Diameter (D_B)	Nominal Ball Diameter	Basic Diameter (D_B)	Nominal Ball Diameter	Basic Diameter (D_B)
	1.0303		1.1865		1.3740
	1.0305		1.1867		1.3742
	1.0307		1.1869		1.3744
	1.0309		1.1871		1.3746
	1.0311		1.1873		1.3748
1-1/32	1.0313	1-3/16	1.1875	1-3/8	1.3750
	1.0315		1.1877		1.3752
	1.0317		1.1879		1.3754
	1.0319		1.1881		1.3756
	1.0321		1.1883		1.3758
	1.0323		1.1885		1.3760
	1.0325		1.1887		1.3762
	1.0327		1.1889		1.3764
	1.0329		1.1891		1.3766
	1.0331		1.1893		1.3768
	1.0333		1.1895		1.3770
	1.0335		1.1897		1.3772
	1.0337		1.1899		1.3774
	1.0615		1.2490		1.4365
	1.0617		1.2492		1.4367
	1.0619		1.2494		1.4369
	1.0621		1.2496		1.4371
	1.0623		1.2498		1.4373
1-1/16	1.0625	1-1/4	1.2500	1-7/16	1.4375
	1.0627		1.2502		1.4377
	1.0629		1.2504		1.4379
	1.0631		1.2506		1.4381
	1.0633		1.2508		1.4383
	1.0635		1.2510		1.4385
	1.0637		1.2512		1.4387
	1.0639		1.2514		1.4389
	1.0641		1.2516		1.4391
	1.0643		1.2518		1.4393
	1.0645		1.2520		1.4395
	1.0647		1.2522		1.4397
	1.0649		1.2524		1.4399
	1.1240		1.3115		1.4990
	1.1242		1.3117		1.4992
	1.1244		1.3119		1.4994
	1.1246		1.3121		1.4996
	1.1248		1.3123		1.4998
1-1/8	1.1250	1-5/16	1.3125	1-1/2	1.5000
	1.1252		1.3127		1.5002
	1.1254		1.3129		1.5004
	1.1256		1.3131		1.5006
	1.1258		1.3133		1.5008
	1.1260		1.3135		1.5010
	1.1262		1.3137		1.5012
	1.1264		1.3139		1.5014
	1.1266		1.3141		1.5016
	1.1268		1.3143		1.5018
	1.1270		1.3145		1.5020
	1.1272		1.3147		1.5022
	1.1274		1.3149		1.5024

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TABLE I. Basic diameter (continued).

Nominal Ball Diameter	Basic Diameter (D_B)	Nominal Ball Diameter	Basic Diameter (D_B)
1-9/16	1.5615	1-5/8	1.6240
	1.5617		1.6242
	1.5619		1.6244
	1.5621		1.6246
	1.5623		1.6248
	1.5625		1.6250
	1.5627		1.6252
	1.5629		1.6254
	1.5631		1.6256
	1.5633		1.6258
	1.5635		1.6260
	1.5637		1.6262
	1.5639		1.6264
	1.5641		1.6266
	1.5643		1.6268
	1.5645		1.6270
	1.5647		1.6272
	1.5649		1.6274

TABLE II. Tolerance for individual balls.

(Tolerance in millionths of an inch)

Allowable Ball Diameter Variation (V_D)	Allowable Deviation From Spherical Form (W)	Allowable Surface Roughness (Arithmetical Average)
10	10	1.0

TABLE III. Tolerance for lots of balls.

(Tolerance in millionths of an inch)

Allowable Lot Diameter Variation	Basic Diameter Tolerance	Allowable Ball Gage Deviation		Container Marking Increment
		High	Low	
20	± 100	+ 50	− 40	10

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P.A. OS Other cust AV, 99	International interest	Title Balls, Bearing, Grade 10, Aircraft-Quality Steel	MILITARY STANDARD MS3226
PROCUREMENT SPECIFICATION MIL-B-1083	SUPERSEDES MS3224 in part		PAGE 5 OF 7

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TABLE IV. Supersedure information. 1/

Cancelled part number	Superseded by	Cancelled part number	Superseded by
MS3224-1007	MS3226-0.1250	MS3224-1036	MS3226-0.7500
MS3224-1009	MS3226-0.1563	MS3224-1037	MS3226-0.7813
MS3224-1011	MS3226-0.1875	MS3224-1038	MS3226-0.8125
MS3224-1012	MS3226-0.2188	MS3224-1040	MS3226-0.8750
MS3224-1014	MS3226-0.2500	MS3224-1041	MS3226-0.9063
MS3224-1016	MS3226-0.2813	MS3224-1042	MS3226-0.9375
MS3224-1018	MS3226-0.3125	MS3224-1044	MS3226-1.0000
MS3224-1019	MS3226-0.3438	MS3224-1045	MS3226-1.0625
MS3224-1021	MS3226-0.3750	MS3224-1046	MS3226-1.1250
MS3224-1023	MS3226-0.4063	MS3224-1047	MS3226-1.1875
MS3224-1024	MS3226-0.4375	MS3224-1048	MS3226-1.2500
MS3224-1026	MS3226-0.4688	MS3224-1049	MS3226-1.3125
MS3224-1028	MS3226-0.5000	MS3224-1050	MS3226-1.3750
MS3224-1029	MS3226-0.5313	MS3224-1051	MS3226-1.4375
MS3224-1030	MS3226-0.5625	MS3224-1052	MS3226-1.5000
MS3224-1031	MS3226-0.5938	MS3224-1053	MS3226-1.5625
MS3224-1032	MS3226-0.6250	MS3224-1054	MS3226-1.6250
MS3224-1033	MS3226-0.6563		
MS3224-1034	MS3226-0.6875		
MS3224-1035	MS3226-0.7188		

1/ See note 2.

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REQUIREMENTS:

1. MATERIAL: AIRCRAFT-QUALITY STEEL, CHEMICAL COMPOSITION UNS T11350 IN ACCORDANCE WITH AMS 6491A AND AS SPECIFIED AS COMPOSITION 13 IN PROCUREMENT DOCUMENT.
2. HARDNESS: AS SPECIFIED IN PROCUREMENT DOCUMENT.
3. SURFACE ROUGHNESS: TOLERANCE LIMITS FOR SURFACE ROUGHNESS AS SPECIFIED IN TABLE II AND IN ACCORDANCE WITH ANSI B46.1.
4. MATERIAL DENSITY: MATERIAL DENSITY SHALL BE 0.279 POUND PER CUBIC INCH.
5. BALL GRADE: ALL BALLS SHALL BE GRADE 10 AS DEFINED BY AFBMA-STD-10.
6. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THIS MS NUMBER FOLLOWED BY A DASH AND THE BASIC DIAMETER REQUIRED AS SELECTED FROM TABLE I.

EXAMPLE: MS3226 - 0.9992

BASIC DIAMETER REQUIRED

THIS MS NUMBER

MS3226-0.9992 INDICATES A GRADE 10 BEARING BALL OF AIRCRAFT-QUALITY STEEL WITH A BASIC DIAMETER OF 0.9992 INCH.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES. COLUMN HEADINGS IN TOLERANCE TABLES ARE IN ACCORDANCE WITH AFBMA-STD-10.
2. SOME MS3226 PART NUMBERS SUPERSEDE SOME MS3224 PART NUMBERS (SEE TABLE IV). MS3224 BALLS CANNOT BE USED AS REPLACEMENTS FOR MS3226 BALLS.
3. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
4. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

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PROCUREMENT SPECIFICATION MIL-B-1083	SUPERSEDES MS3224 in part		PAGE 7 OF 7

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